ATMANIRBHAR BHARAT // DEFENCE STRATEGIC BRIEFING

HEATBOLT

SOVEREIGN PORTABLE THERMAL INFRASTRUCTURE FOR EXTREME ENVIRONMENTS

An indigenous, water-activated thermal platform engineered to deliver instant heat and steam without mains electricity, LPG cylinders, or open flame.
Engineered for high-altitude border supremacy (Siachen / Eastern Ladakh / LAC), combat casualty preservation, and frontline soldier survival where conventional infrastructure cannot follow.

DR. KUMARPAL PORWAL — FOUNDER & MANAGING DIRECTOR
+91 73533 19757 | founder.heatbolt@gmail.com | Bangalore, India
Defence Innovation & Technology Commercialisation

MINISTRY OF DEFENCE & ARMED FORCES BRIEFING
High-Altitude & Winter Warfare Doctrine
100% Make-in-India Intellectual Property

Thermal Infrastructure at the Point of Need

// STRATEGIC DEFENCE DOCTRINE

Move thermal infrastructure from the base to the soldier.

Heatbolt is developing an indigenous, water-activated thermal platform providing controlled heat and steam without dependence on fixed infrastructure — designed for frontline soldier warmth, hot ration hydration, high-altitude alpine warfare, combat casualty support, and forward shelters.

- Soldier Warmth
- Hot Combat Rations
- High-Altitude Ops
- Snow-to-Water Conversion
- Combat Casualty Care
- Field Shelters
- Hygiene & Sanitation
- Weapon De-Icing
- UAV Subsystems
- Search & Rescue

SOVEREIGN DEFENCE VALUE — Every frontline soldier must possess dependable thermal independence. Frontline endurance, physical combat agility, cognitive acuity, and casualty survival in -40°C glacial theatres depend directly on immediate, localized thermal energy.

Extreme Environments Create a Thermal Infrastructure Gap

THE BORDER & HIGH-ALTITUDE LOGISTICS BOTTLENECK

A soldier in high-altitude, remote, glacial, winter or infrastructure-constrained terrain needs thermal energy for far more than personal warmth.

Modern military logistics still rely heavily on heavy diesel generators, vulnerable fuel bladders, and volatile kerosene stoves for forward mountain positions. When patrol detachments deploy to isolated ridges, observation posts, and glacial passes, external infrastructure is virtually non-existent.

Without portable, flameless, solid-state thermal capability, hypothermia rates spike, combat rations freeze into unpalatable blocks, medical triage is severely compromised, and weapon linkages seize in critical moments.

Eight Thermal Requirements. Zero Infrastructure.

DEMAND #01

01 Personnel Warmth

Sustained warmth, frostbite prevention, survival, and cold-weather physical endurance.

DEMAND #02

02 Combat Rations

Flameless heating of RTE / RTC rations, hot tea, and vital warm hydration on patrol.

DEMAND #03

03 Water Utility

Melting high-altitude snowpack into hot drinking water and utility steam.

DEMAND #04

04 Field Medical Care

Casualty hypothermia prevention during evacuation and temperature-controlled medical triage.

DEMAND #05

05 Equipment Readiness

Drying tactical gear, unfreezing weapon breach blocks, de-icing optics, and vehicle start.

DEMAND #06

06 Shelter Heating

Temporary heating of alpine bivouac shelters, tents, and forward observation posts.

DEMAND #07

07 Tactical Mobility

Cold-weather pre-heating for vehicle engine blocks, optics, and UAV lithium batteries.

DEMAND #08

08 Emergency SAR

High-altitude search & rescue, avalanche survivor extraction, and disaster relief.

The Challenge of Useful Heat

THE FUNDAMENTAL COMBAT PARADOX

**“The challenge is not simply producing heat.
It is producing USEFUL HEAT where conventional infrastructure cannot easily follow.”**

Sub-Zero (-40°C)

Remote Mountain Ridge

Blackout Operations

Zero Logistical Trail

Physically Demanding

Zero Carbon Monoxide

THE FRONTLINE SOLDIER AS THE BENCHMARK FOR EXTREME ENGINEERING

The frontline soldier operating in glacial altitude represents the ultimate operating stress test. If a thermal platform performs reliably under high-altitude, wind-chill, and sub-zero conditions, every adjacent commercial, industrial, and humanitarian application becomes a natural expansion — not an unvalidated risk.

Portable Thermal Infrastructure

100% INDIGENOUS REACTION ENGINEERING

A proprietary water-activated exothermic system — engineered specifically for extreme environments where infrastructure is zero.

Heatbolt eliminates dangerous open fires, heavy generators, and hazardous pressurized fuel canisters through a solid-state, highly stable mineral formulation.

Activated by ambient water, stream runoff, or melted snowpack, Heatbolt converts chemical bond energy directly into rapid, controlled thermodynamic heat and clean steam in seconds — with 100% indigenous Indian raw materials.

Water In. Heat Out. Anywhere.

01

ACTIVATE

Water or melted snow is introduced into the hermetic thermal pouch chamber.

02

REACT

Proprietary mineral formulation triggers an instant, controlled exothermic reaction.

03

TRANSFER

Thermal energy and pure steam conduct precisely into the payload without contamination.

04

DEPLOY

Engineered into soldier pouches, ration packs, survival vests, and tactical equipment.

Zero Mains Power

Zero LPG / Gas Need

Zero Open Flame

Zero Acoustic Signature

Zero Toxic Residue

Modular Form Factors

TACTICAL BLACKOUT DISCIPLINE — Flameless chemical activation emits zero light signature and zero acoustic profile, fully conforming to strict night blackout and forward sentry concealment doctrines.

Not a Heater. A Thermal Platform.

A conventional product solves one problem. A platform solves many — using one common technological core.

DOMAIN #01

Soldier Warming

Personal thermal protection at point of need; wearable heating elements.

DOMAIN #02

Ration Heating

Hot meals and beverages without field kitchens, flame, or combustion smoke.

DOMAIN #03

Hot Water / Steam

Field utility hot water generation, snow-to-water conversion, sanitation.

DOMAIN #04

Temporary Heating

Bivouac shelters, observation posts, forward operational living pods.

DOMAIN #05

Field Medical

Thermal workflows for casualty care, hypothermia treatment, IV warming.

DOMAIN #06

Equipment Drying

Tactical gear drying, weapon de-icing, moisture mitigation in freezing zones.

DOMAIN #07

Cold-Weather Mobility

Battery pre-heating, UAV sub-systems, vehicle component readiness.

DOMAIN #08

Emergency SAR

Disaster response, avalanche search & rescue, humanitarian mission thermal kits.

Selected Figures from Current Development

100°C

Peak Thermal Output

Selected high-performance configurations

< 30s

Activation Latency

Approximate rapid-response trigger

45 min

Active Thermal Duration

Standard military operational envelope

60°C / 3h

Extended Warmth Profile

Casualty transport & survival configuration

QUALIFICATION MANDATE — Performance figures reflect developmental laboratory envelopes. Output kinetics vary with formulation density, water volume, packaging geometry, payload thermal conductivity, ambient sub-zero temperature, and wind chill. Every Defence application undergoes rigorous mil-spec qualification.

From Product to Soldier Capability

MISSION-ALIGNED DEFENCE SOLUTIONS

Ten mission-aligned use cases — each engineered toward rigorous military qualification, not mere laboratory demonstration.

Heatbolt translates fundamental chemical thermodynamics into practical soldier-centric kits.

From extreme glacial combat at -40°C on the Siachen glacier to rapid forward triage casualty evacuation, Heatbolt creates modular capability that integrates directly into existing defence procurement specifications.

Ten Use Cases. One Core Platform.

USE CASE 01

UC-01 Personal Thermal

Body warmers, survival packs, heated gloves/jackets for prolonged sub-zero sentry watch.

USE CASE 02

UC-02 Field Food & Beverages

Flameless self-heating of combat rations (RTE/RTC meals), tea, coffee, and clean hot water.

USE CASE 03

UC-03 High-Altitude Glacial

Warming, hot hydration, shelter heating, gear drying, and continuous snow melting.

USE CASE 04

UC-04 Ice & Snow Management

Localised melting, clearing frozen weapon mounts, optics, and locked access-points.

USE CASE 05

UC-05 Field Medical Support

Casualty transport hypothermia prevention, warm IV fluids, and field sterilisation.

USE CASE 06

UC-06 Temporary Shelters

Rapid emergency heating for tactical bivouacs, alpine shelters, and forward command posts.

USE CASE 07

UC-07 Hygiene & Sanitation

On-demand hot water, steam generation, field laundry, and decontamination stations.

USE CASE 08

UC-08 Equipment Maintenance

Weapon bore drying, vehicle engine block thawing, and frozen linkage recovery.

USE CASE 09

UC-09 Aviation & UAV Systems

Pre-flight surface de-icing and cold-weather UAV battery pre-heating support.

USE CASE 10

UC-10 Search & Rescue (SAR)

High-altitude disaster recovery, avalanche rescue thermal bags, and multi-agency response.

A Family of Mission-Ready Kits

KIT 01

Soldier Survival Kit

CORE CAPABILITIES:

Portable thermal activation • Emergency body warming • Food & beverage heating • Clean hot water

MISSION PROFILE:

Immediate survival loadout for extreme weather patrols and covert reconnaissance detachments.

KIT 02

High-Altitude Kit

CORE CAPABILITIES:

Personal warmth modules • Snow / ice thawing packs • Weapon / gear drying • Bivouac shelter heat

MISSION PROFILE:

Engineered for 15,000+ ft glacial environments, permanent snowlines, and sub-zero sentry posts.

KIT 03

Field Ration System

CORE CAPABILITIES:

Integrated heating sleeve • Hot meals in 10 mins • Hot hydration packs • Controlled steam venting

MISSION PROFILE:

Self-contained combat meal heating system replacing flame stoves and unpalatable frozen food.

KIT 04

Winter Operations Kit

CORE CAPABILITIES:

Extended body warming • Heavy gear drying • Utility hot water • Mobile field clinic heating

MISSION PROFILE:

Comprehensive squad-level thermal kit for extreme winter warfare and prolonged field maneuvers.

High-Altitude & Glacial Operations

GLACIAL THEATRE CAPABILITIES

- **Soldier Sustained Warming**
  Continuous personal thermal protection in -40°C blizzards; prevents frostbite and peripheral vasoconstriction.
- **Hot Food & Instant Hydration**
  Rapid heating of canned and pouch rations without stoves; eliminates reliance on frozen solid rations.
- **Shelter & Bivouac Heating**
  Safe, non-toxic flameless warmth for alpine tents without dangerous carbon monoxide buildup.
- **Snow & Ice Conversion**
  Converts glacial snow into warm potable drinking water in minutes; prevents severe dehydration at altitude.

TARGET THEATRES

Siachen Glacier • Eastern Ladakh • Tawang Sector • Drass & Kargil Passes • High-Altitude Forward Posts

WHY HIGH-ALTITUDE LEADS THE STRATEGY:

“High-altitude operations represent the harshest thermal infrastructure gap in modern Defence.

If Heatbolt performs at 18,000 feet under -40°C extreme winds, every adjacent ground environment becomes an operational expansion — not an unproven risk.”

Combat Medical Support — Qualification Discipline

TRIAGE CRITICAL

Patient Evacuation Care

Continuous thermal support during casualty evacuation (CASEVAC); prevents hypothermic coagulopathy in wounded personnel.

UTILITY SUPPORT

Hot Water & Steam Utility

Immediate hot water for trauma cleaning, patient re-warming baths, and medical kit reconstitution.

FIELD STERILISATION

Sterilisation Concepts

Flameless high-temperature steam generation for field instrument sterilisation and emergency decontamination workflows.

CLINICAL REGULATORY DISCIPLINE — Heatbolt does not propose bypassing medical qualification protocols. Dedicated clinical evaluation programs cover precise thermal stability, biocompatibility, non-toxic emissions, zero volatile organic compounds, human factors, and regulatory approvals. Medical applications represent specialized development tracks.

Aviation, UAV & Emergency Response

AEROSPACE HORIZONS

Aviation & UAV — Future Concepts

Exploratory applications focusing on cold-weather aviation readiness:

• Aircraft De-Icing Support: Rapid thermal surface thawing for rotary blades and wings.
• UAV Battery Pre-Heating: Lithium-ion battery conditioning to restore operational flight time in sub-zero air.
• Subsystem Warming: Thermal management for high-altitude sensor gimbals and forward-looking infrared optics.
• Field Maintenance: Rapid thawing of seized linkages and hydraulic junctions.

DUAL-USE HUMANITARIAN

Search & Rescue — Dual-Use Value

One unified platform serving Defence, Disaster Response, and Humanitarian Relief:

• Rapid Casualty Warming: Rapid thermal blankets for hypothermic survivors.
• Emergency Shelters: Instant heat for avalanche rescue camps and disaster survivor zones.
• Dual-Use National Framework: Deployment across Indian Army, NDRF, SDRF, and global humanitarian agencies.
• Sovereign Resilience: Stockpiled emergency thermal reserves requiring zero maintenance or batteries.

One Technology. 80+ Application Concepts.

A structured innovation pipeline across 9 operational domains — representing structured R&D horizons, not premature claims.

DOMAIN 01

Personnel

Body warmers • Warmer jackets • Survival packs • Portable baths • Field laundry

DOMAIN 02

Food & Nutrition

Field kitchens • Food warming • Preservation • Field bakery • Self-heating rations

DOMAIN 03

Water & Steam

Water heating • Steam generation • Distillation • Water purification concepts

DOMAIN 04

Medical & Triage

Field hospitals • Casualty warming • Instrument sterilisation • Trauma kits

DOMAIN 05

Equipment & Ops

Gear drying • Vehicle maintenance • Weapon bore cleaning • De-icing systems

DOMAIN 06

Infrastructure

Concrete curing in cold • Remote building • Emergency shelters • Underground posts

DOMAIN 07

Security & Border

Border post warming • Surveillance camera de-fogging • Tactical fogging concepts

DOMAIN 08

Search & Rescue

Disaster response • Avalanche survivor kits • Humanitarian emergency supply

DOMAIN 09

Environmental

Environmental decontamination • Waste treatment thermal support • Remote field ops

Technology Alone Does Not Create a Defence Capability

EXECUTION & SCALE

Heatbolt is assembling sovereign capability across every dimension — from atomic reaction kinetics to millions of consistent field units.

A breakthrough technology is only as viable as its manufacturing repeatability, supply chain resilience, and institutional leadership.

Heatbolt brings together seasoned industrial titans, aerospace propulsion veterans, FMCG packaging leaders, and defence strategists to bridge the gap between invention and multi-million unit military deployment.

Built to Execute

FOUNDER & MANAGING DIRECTOR

Dr. Kumarpal Porwal

Deep-tech commercialisation • Controlled thermal activation • Enterprise AI & operational transformation (Chief of Staff, Vega Aviation) • Defence & FMCG strategic engagement.

MANUFACTURING & QUALITY HEAD

Rahul Awasthi

National Mfg & Quality Head, SBC Tanzania (Pepsi) • Ex-SLMG Coca-Cola, Amazon (led 6,000+ operations workforce), Glenmark, SABMiller, Coca-Cola.

MARKETING & GROWTH

Puneet Solanki

Market positioning & institutional engagement • Application & partnership development • Go-to-market scaling across Defence, institutional buyers, and global consumer brands.

GLOBAL BUSINESS LEADERSHIP

Amit Jain

Former Managing Director — L'Oréal India (First Indian MD) • Executive leadership across AkzoNobel, Coca-Cola, Viacom/MTV, ICI • Consumer innovation & global scaling.

A Multidisciplinary Research Bench

Dr. Saibalendu Sarkar — TIFR [NANOMATERIALS & CATALYSIS]
Nanomaterials, reaction kinetics, CO2 conversion, advanced mineral characterisation and thermodynamic profiling.

Anil Kumar Tripathi [PACKAGING & BARRIER ENGINEERING]
Packaging technology, high-barrier hermetic foils, QA/QC standards, shelf-life integrity, and new product development.

Mahantesh G. Vantamuri [MECHANICAL & TOOLING DESIGN]
Mechanical tooling, product industrial design, CAD/CAM simulation, injection moulding, and high-speed automated assembly.

T. Koshi • Prerna Bhanot • Michele Barbazza [STRATEGY, TALENT & EXPANSION]
Corporate governance, institutional talent scaling, international business development, and sovereign trade frameworks.

Extended Institutional Network [DEFENCE & FOOD TECH ECOSYSTEM]
Ex-CFTRI scientists, All-India food industry leadership, Vega Helmets industrial network, Microlabs, Nilons, and advanced research labs.

Names That Strengthen the Mission

ROCKET SCIENCE & PROPULSION

Nambi Narayan

Legendary ISRO Rocket Scientist — Space propulsion pioneer, national strategic aerospace leadership.

PACKAGING DEVELOPMENT & FMCG SCALE

Pradeep Kumar Pandey

Senior Advisor — President, Packaging Development, Emami Ltd • 29+ years, Gold Medalist • Ex-Procter & Gamble, Amway, Dabur.

GLOBAL THERMOFORMING & PACKAGING

Andrew Dobringer

Senior Advisor — International Thermoforming, Precision Tooling & Advanced Packaging Systems Engineering Specialist.

RESEARCH LEADERSHIP & MATERIALS

Prof. Somnath Ganguly

Senior Advisor — 40+ years of distinguished leadership in chemical formulation, materials research, and laboratory scaling.

MACHINERY & AUTOMATION

Ravi Tejwani

Industrial machinery expert — Manufacturing automation, precision production equipment, and high-speed packaging lines.

NARRATIVE & STRATEGIC ENGAGEMENT

R. Madhavan

Acclaimed film actor & director — National communication, strategic storytelling, and high-impact institutional narrative.

STRATEGIC ENTERPRISE NETWORK

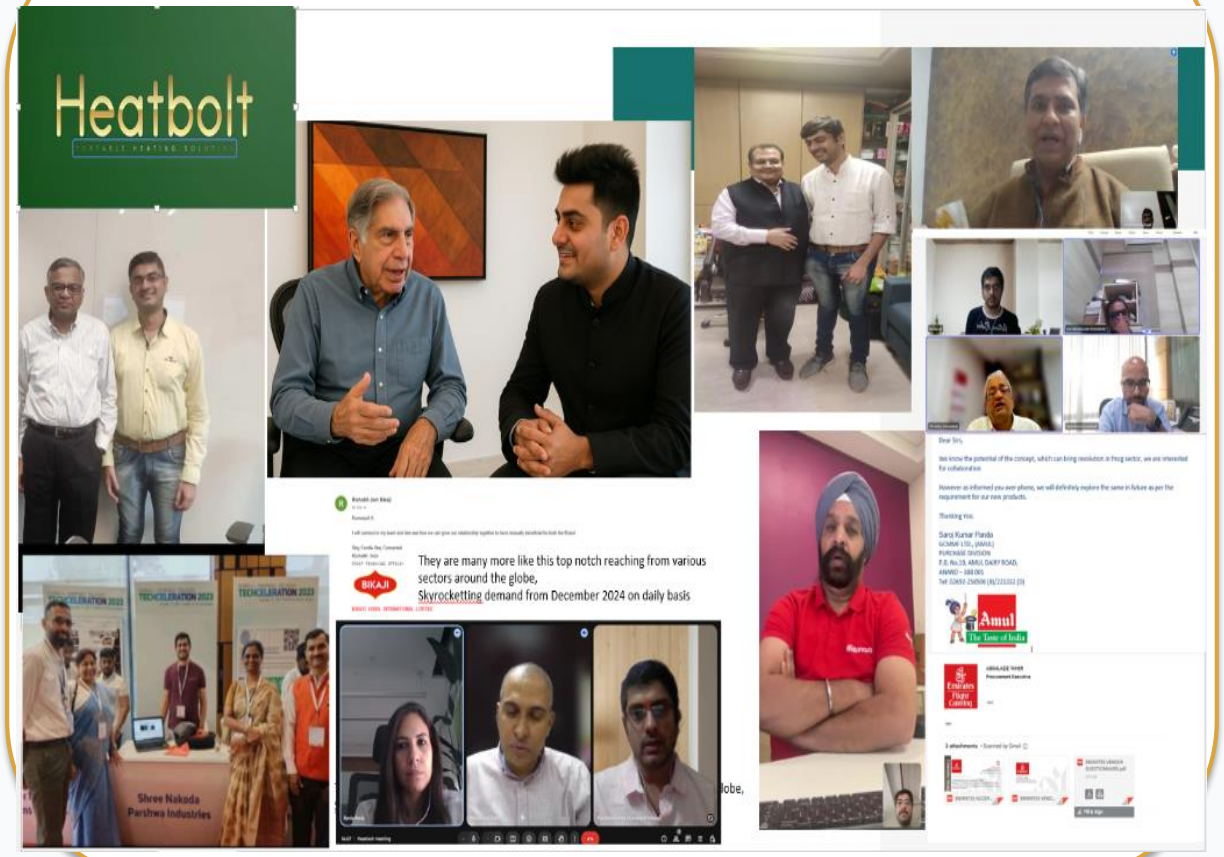
Marichi Mathur

Senior Advisor — Strategic growth, enterprise leadership, and national defence institutional network expansion.

GOVERNANCE NOTE — Advisory engagements are presented according to formal domain areas and institutional supporting documentation.

National Titans, AMUL & Global Aviation Validation

EVIDENCE FILE: SHRI RATAN TATA CONSULTATION • AMUL LETTER • EMIRATES CATERING



PATRIARCHAL ENDORSEMENT

Shri Ratan N. Tata Direct Consultation

Personal strategic consultation with India's legendary industrial patriarch, reviewing Heatbolt's self-heating technology for sovereign defence resilience and national grassroots impact.

NATIONAL COOPERATIVE GIANT

AMUL (GCMMF Ltd) Official Collaboration Request

Official written communication from the Purchase Division: 'We know the potential of the concept, which can bring revolution in FMCG sector, we are interested for collaboration...' — India's largest food cooperative.

GLOBAL AVIATION SUPPLY

Emirates Flight Catering Onboarding

Formal international airline vendor qualification and commercial onboarding for in-flight self-heating meals and expeditionary food solutions.

SURGING MARKET PULL

Surging Cross-Sector National Inbound Demand

Active high-level engagements with Bikaji Foods CFO, Tier-1 defence suppliers, and major packaging summits (Techceleration 2023 / Startup India) projecting 20B+ annual unit volume.

VERIFIED INDUSTRY ADOPTION — Documented traction proves Heatbolt has crossed the credibility threshold: backed by India's foremost industrial leaders, national FMCG cooperatives, and international aviation supply chains.

The Ecosystem Is Already Responding

DEFENCE VALIDATION

Institutional Defence Pull

Active engagement across Indian Armed Forces command hubs, DRDO ecosystem partners, and emergency response teams for high-altitude cold trials.

FMCG ENGAGEMENT

Global FMCG Traction

Direct commercial, self-heating ration, and meal packaging evaluations initiated with Nestlé, Barilla, Mondelēz, PepsiCo, Unilever, and McDonald's.

SCALE TRACTION

Commercial Momentum

Showcased via Startup India and Invest India platforms; over 60,000+ stakeholder touchpoints and 20B+ annual unit volume potential projected.

PROVEN DEMAND INDICATORS — Heatbolt is transitioning rapidly from breakthrough laboratory formulation to mission-critical defence field evaluation and high-volume industrial packaging. Tier-1 multinational food conglomerates and defence forces have validated the immense market gap for portable, flameless, water-activated thermal energy.

Path to Operational Deployment

PHASE 01

Field Trials

OPERATIONAL TARGET:

Sub-Zero Stress Testing

MILESTONES:

Rigorous sub-zero & high-altitude environmental stress testing with designated frontline infantry units.

PHASE 02

Mil-Spec Certification

OPERATIONAL TARGET:

Standards Qualification

MILESTONES:

Thermal output standardization, shelf-life validation (3+ yrs), and hazardous transport compliance.

PHASE 03

Volume Production

OPERATIONAL TARGET:

Automated Packaging

MILESTONES:

Automated pouch filling and QC scaling with Tier-1 manufacturing partners for 10M+ units monthly.

PHASE 04

Force Integration

OPERATIONAL TARGET:

Standard Issue Deployment

MILESTONES:

Standard issue inclusion into ration kits, extreme climate survival packs, and mobile field clinics.

JOIN THE MISSION // SOVEREIGN DEFENCE ENGAGEMENT

HEATBOLT

Partnering to Equip the Frontline Soldier with Sovereign Thermal Independence

"The future of frontline combat endurance and extreme survival is not just portable food and water — it is sovereign portable heat."

DR. KUMARPAL PORWAL — FOUNDER & MD

+91 73533 19757 | +91 88672 94189

founder.heatbolt@gmail.com | Bangalore, India

DEFENCE & STRATEGIC ENGAGEMENT

MIL-SPEC TRIALS (NORTHERN COMMAND / DRDO / iDEX)

TECHNOLOGY LICENSING & IP DEFENCE INTEGRATION

NATIONAL STRATEGIC INDUSTRIAL ALLIANCES